Mind over Matter:

Attentional Hold and Capture Measured by Event-Related Potentials

Stella R. Schick

Department of Psychology & Neuroscience, The University of North Carolina at Chapel Hill

PSYC 395: Independent Research

Dr. Joseph B. Hopfinger

April 21, 2025

Mind over Matter:

Attentional Hold and Capture Measured by Event-Related Potentials

Out of the many aspects of human cognition, visual attention plays one of the most important roles, allowing people to process and respond to stimuli (Hopfinger et al., 2000). Electroencephalography (EEG) quantitatively measures the electrical impulses behind cognitive processes (Luck & Kappenman, 2012). Its high temporal specificity creates continuous reports of brain waves, providing researchers with an accurate and accessible method to monitor levels of visual attention (Wójcik et al., 2018). EEG data can be segmented into event-related potential (ERP) components (Luck, 2014), each of which represents a specific neural process; though many of these components correspond to attention, this paper will focus on the sustained posterior contralateral negativity (SPCN) and N2pc and their role in attentional hold and capture.

Like many ERP components that relate to visual processing, the N2pc is a negative deflection that appears in the parietal and occipital scalp areas, typically appearing around 200-300 ms after stimulus presentation (Luck, 1993; Luck & Hillyard, 1994). It is observed in the parietal and occipital areas of the brain, always contralateral to the visual stimulus. Stimuli in the lower visual field elicit a significantly stronger N2pc than stimuli in the upper visual field (Chen et al., 2022; Dolci et al., 2024). In EEG recordings, the N2pc is strongest around electrodes PO7 and PO8, since these electrodes best measure the parietal and occipital sites (Eimer & Kiss, 2007). Several other components that often appear alongside the N2pc, notably the SPCN and the response-locked posterior contralateral negativity (RLpcN), are also observed at PO7 and PO8.

The SPCN, like the N2pc, is a negative ERP component seen in the parietal and occipital areas (Jolicœur et al., 2006). Notably, its amplitude is positively correlated with the number of visual stimuli (Chen et al., 2022). It is typically observed at around 400 ms in the hemisphere contralateral to the stimulus: if an image was presented on the right side of a participant's visual field, the SPCN would appear in the left hemisphere of their brain. The third component discussed in this paper, the RLpcN, is a more recently defined component. The RLpcN was proposed by Drisdelle and Jolicœur in a 2019 study, which identified the component as a relative of the SPCN. It appears while the participant prepares their response after viewing the stimulus, starting roughly 700 ms before the response is submitted (Drisdelle & Jolicœur, 2019). Consequently, its duration directly reflects the participant's response times. While the onset of the SPCN correlates with stimulus presentation, the onset of the RLpcN corresponds to response preparation. Temporally, the SPCN lies between the N2pc and the RLpcN, with the RLpcN being the closest to the response and the N2pc being the furthest. All three of these components are derived from contralateral mins ipsilateral difference waves, since the neural processes they relate to are strongest at the contralateral sites on the scalp (Luck, 2014). Subtracting the ipsilateral waves from the contralateral waves reduces any background neural noise and isolates the desired ERP component (Luck, 2014).

Literature Review

Many current attentional studies focus on the SPCN and N2pc. Chen et al. centered their experiment (2022) around the effects of stimulus placement on the amplitude of the N2pc and SPCN. They noted that, in prior research, a bilateral N2pc had been evoked by stimuli on the vertical midline, the line that separates the right and left sides of the screen (the horizontal midline, conversely, separates the top and bottom halves of the screen). Though the effects of midline stimulus placement on the N2pc had been studied, no research had yet been done on the equivalent effects on the SPCN. In this study, they aimed to measure how stimuli on the vertical and horizontal midlines affect the N2pc and the SPCN. They predicted that a bilateral SPCN and N2pc would be seen in response to lateral and vertical stimuli, and expected to see an overlap between SPCN and N2pc responses to similar stimuli. Lastly, they hypothesized that the SPCN, like the N2pc, would show a greater response to stimuli in the lower visual field.

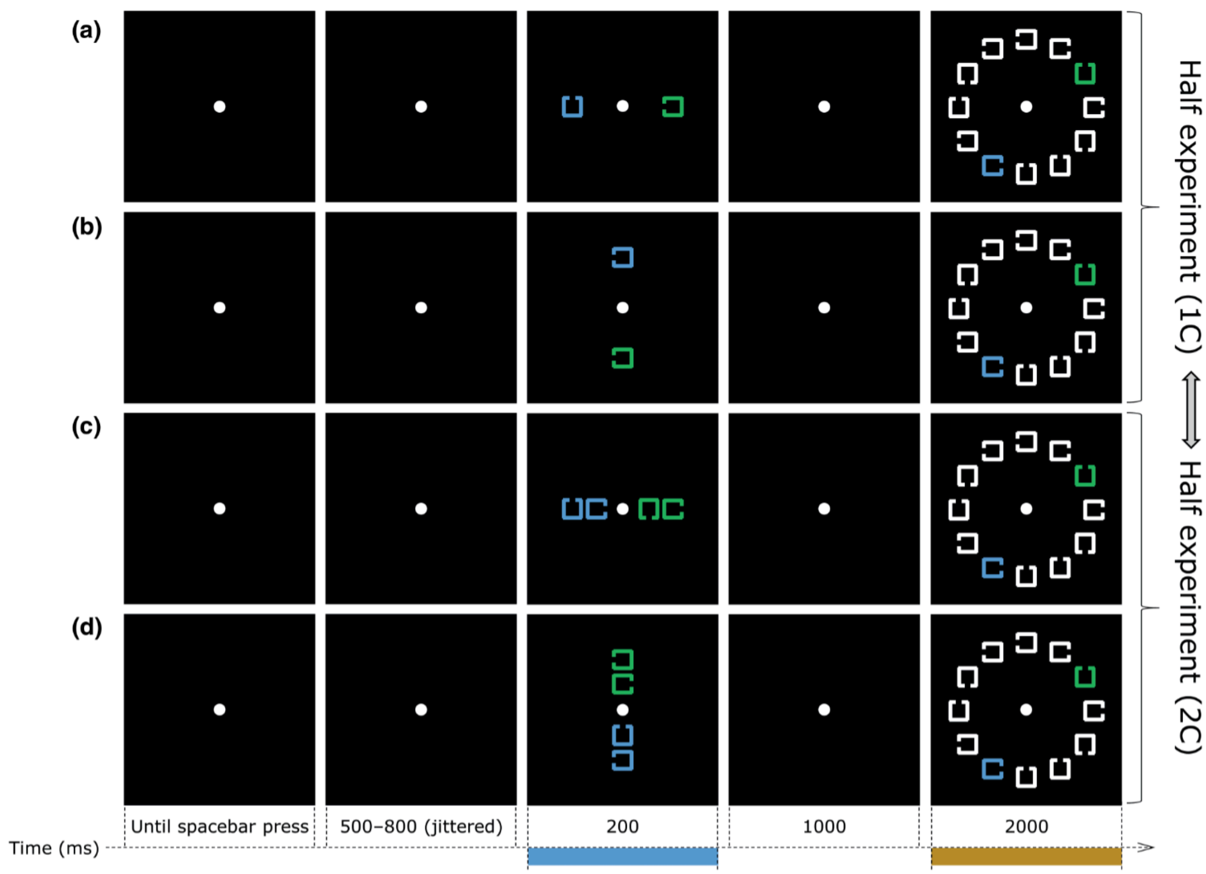

Figure 1. The visual search task used by Chen et al. (2022)

Participants (n = 19) completed a visual task split into two experiments, one requiring less cognitive effort and one requiring more (Chen et al. 2022). Across both experiments, stimuli were notched squares that varied in color (either green or blue) and appeared on a black background (see Figure 1). Stimuli were presented twice and interspersed with a fixation dot. The first round of stimuli were presented on either the vertical or horizontal midline, and the second round of stimuli were grouped in a circle around the fixation dot. In the first experiment, there was one target was present in the initial set of stimuli, while in the second, there were two. Within the second set of stimuli, the target would only reappear in 50% of trials; participants responded to indicate whether the target was present or not. If the target was blue, the first distractors would be green, and vice versa, while the second distractors were always white.

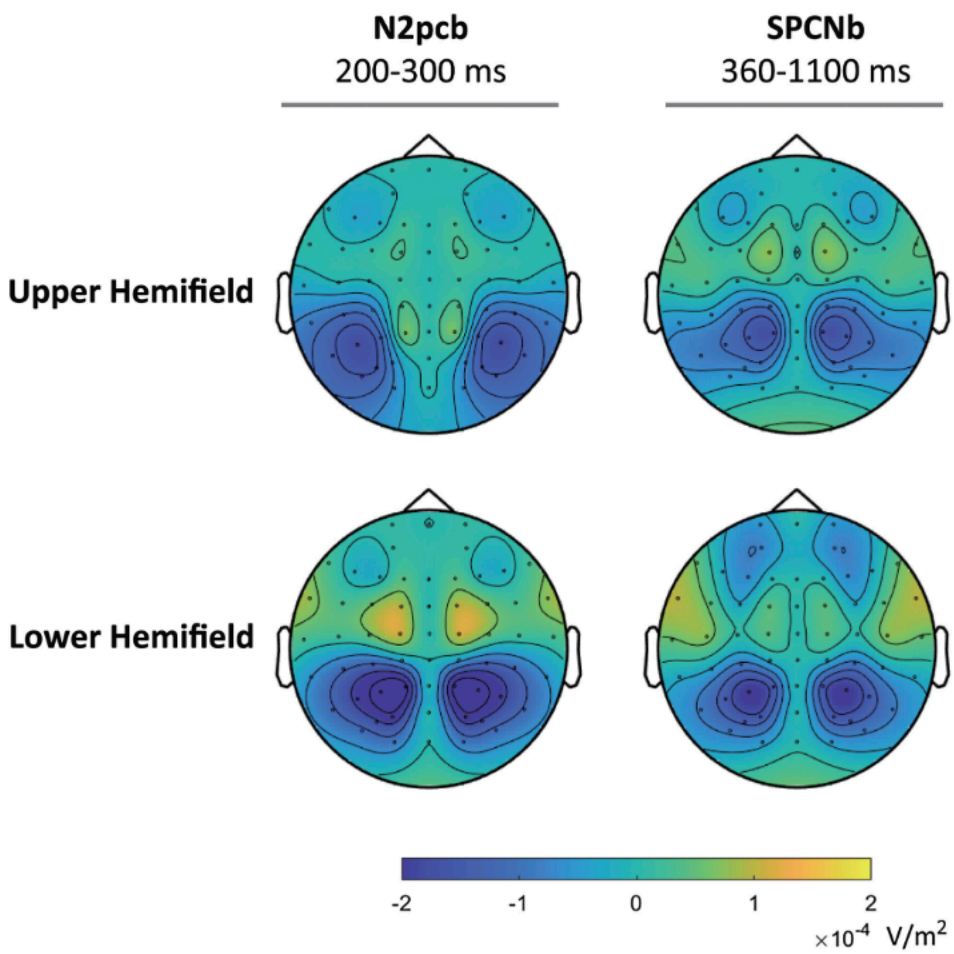

Figure 2. Scalp maps of EEG data in Chen et al. (2022).

The results of their data analysis showed that response times were both shorter and more accurate in the first experiment than in the second. Accuracy was higher with lateral stimuli as opposed to midline stimuli, and in both components, a greater amplitude was seen following stimuli in the lower visual field. In the second experiment, the amplitude of a subcomponent, N2pcb, was greater than that of the N2pc. An equivalent subcomponent of the SPCN, the SPCNb, was measured, but it was found to be statistically equivalent to the main component. Notably, the SPCN was amplified as the number of cues increased, correlating with the greater cognitive load. Chen et al. (2022) demonstrated the overlap between the N2pc and SPCN (see Figure 2), both of which showed a preference for stimuli on or below the lateral midline.

Drisdelle and Jolicœur (2019), the first to propose the RLpcN, designed an experiment measuring all three components. Since the RLpcN occurs while the participant responds to stimuli, they expected to see its duration directly correlating with response times. Additionally, because the SPCN and the RLpcN are closely related, Drisdelle and Jolicœur predicted that the components would exhibit the same behaviors. They hypothesized that RLpcN variance is correlated with attention, processing, and response time. They also predicted that, with shorter response times, the amplitude of the N2pc would be larger and onset latency would be earlier; this prediction was confirmed in their results. Specific to the RLpcN, they hypothesized that, with longer response times, the component would begin earlier.

Participants (n = 154) completed a typical visual search task, where stimuli were notched squares, two presented on either side of the fixation cross (shown in Figure 3). The target square could be either green or orange, though orange was more frequent, while the distractors squares were all blue. Notches could appear on any side of the squares, but notches on the left, right, and

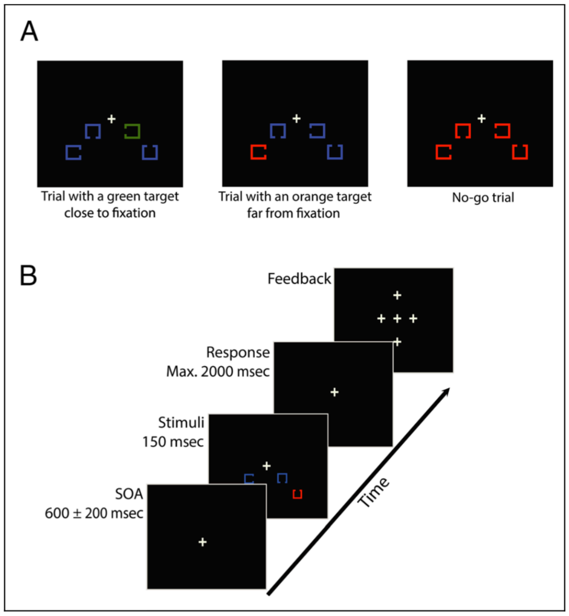

Figure 3. The visual search task used by Drisdelle & Jolicœur (2019).

bottom sides were the most frequent (75%), while top-notched squares were the least frequent (25%). Participants received feedback on their performance throughout the experiment. Analysis of the EEG data (see Figures 4, 5, and 6) showed that the amplitude of the N2pc was positively correlated with the side-notched squares, targets closer to the fixation, and shorter response times. The SPCN, comparatively, occurred earlier and showed a greater amplitude after top-notched squares. The component's onset was also earlier following fast trials, and its amplitude was much larger following targets closer to the fixation cross. As predicted, the RLpcN, like the SPCN, was larger after top-notched squares and targets closer to fixation. Additionally, its onset was correlated with response times (Drisdelle & Jolicœur, 2019). Variance in the onset of the SPCN and RLpcN constitutes a large part of the overlap between stimulus- and response-locked activity.

The results of Drisdelle and Jolicœur's study indicated a substantial overlap between the components, leading them to theorize that the RLpcN could encompass both the N2pc and SPCN (Drisdelle & Jolicœur, 2019). A 2023 study, however, reinforced the idea of the RLpcN as a separate component (Rashal et al., 2023). By measuring both the SPCN and the RLpcN, the study examined visual attention and post-selectional processes. A year prior, a previous iteration of the experiment had focused on the N2pc and contingent negative variation (CNV) instead. The

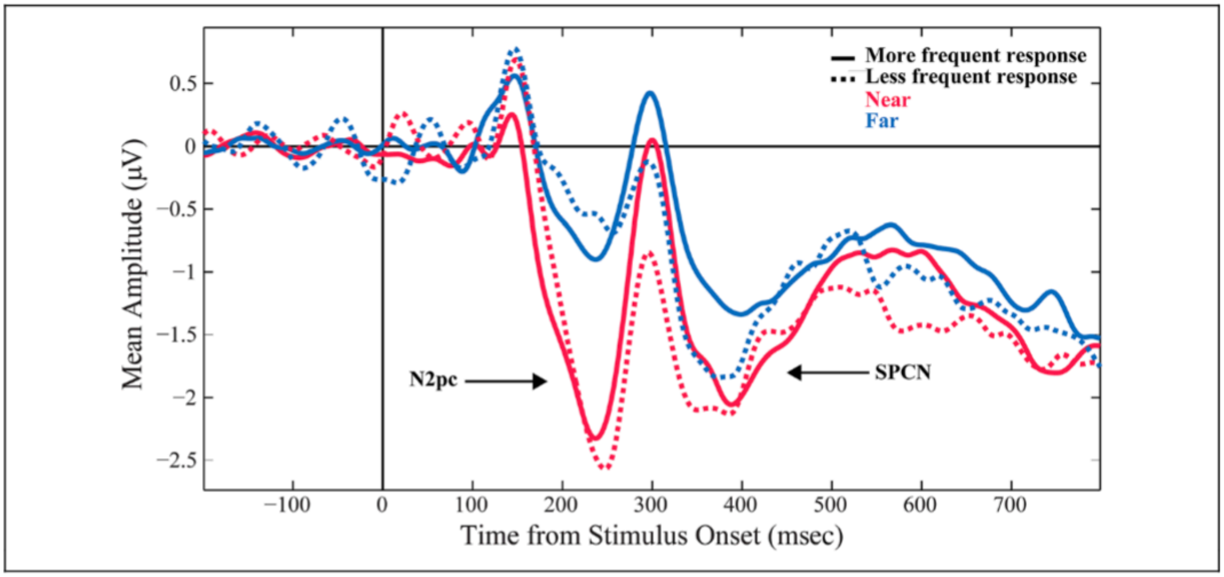

Figure 4. EEG waveforms showing the N2pc and SPCN in Drisdelle & Jolicœur (2019).

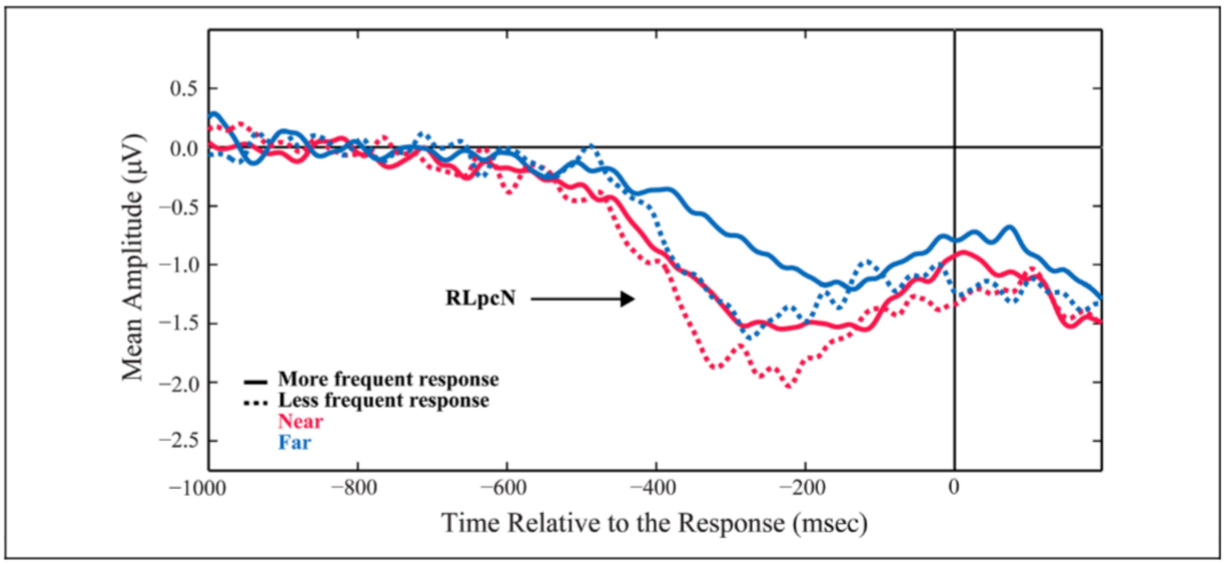

Figure 5. EEG waveforms showing the RLpcN in Drisdelle & Jolicœur (2019).

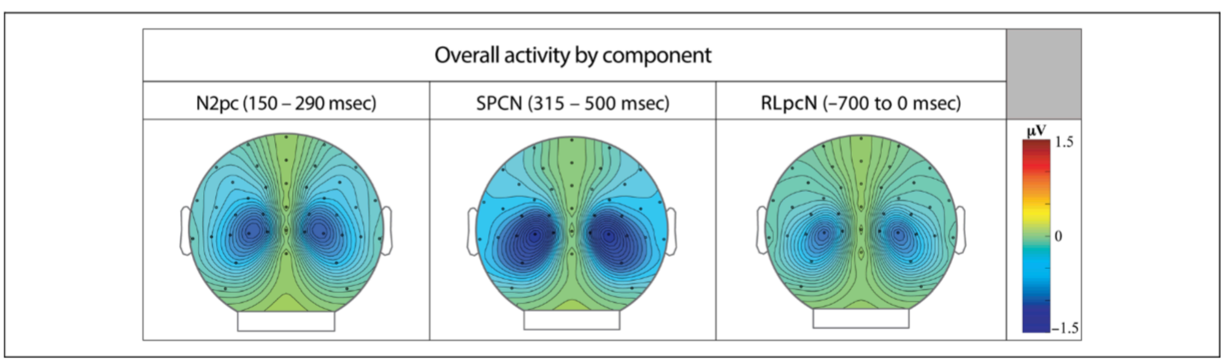

Figure 6. Scalp maps of EEG data in Drisdelle & Jolicœur (2019).

CNV, discovered in 1964 by Walter et al., was the first formally reported ERP component, which, like the RLpcN, was linked to response preparation (Luck, 2014). Rashal et al. (2022) hypothesized that varying task difficulty would affect the two components. To measure this, they divided their task into two similar experiments to compare the effects of difficulty on the N2pc and CNV. They predicted that valid cues would evoke a larger CNV, and salient distractors would produce an N2pc in the more difficult experiment and a P_D in the easier experiment. The P_D, first reported by Hickey et al. (2009), is a contralateral positive deflection that appears concurrently with the N2pc (i.e., from 200-300 ms).

The first experiment that participants performed resembled a non-traditional visual search task, with colored bars as stimuli (shown in Figure 7). One of these bars, the target, was always rotated 20º, while all the other bars were either horizontal or vertical. Participants were tasked with identifying the orientation of the target bar (either clockwise or counter-clockwise) and responding accordingly. In the second experiment, though the target bar was still rotated 20º, the other bars were also diagonal, at 25º. This made distinguishing the target bar significantly more difficult in the second experiment. Additionally, instead of responding based on the orientation, the second experiment required participants to locate a small notch on the target bar. With a presentation of only 300 ms, a substantial amount of focus was needed to correctly respond to each stimulus.

Besides the general setup, several other factors were normalized across the two experiments. Both experiments included four conditions for stimulus presentation. Within the salient-target condition, the target bar would be a different color than the three distractors. Within the no-distractor condition, all four bars were the same color. In the lateral-target condition, one

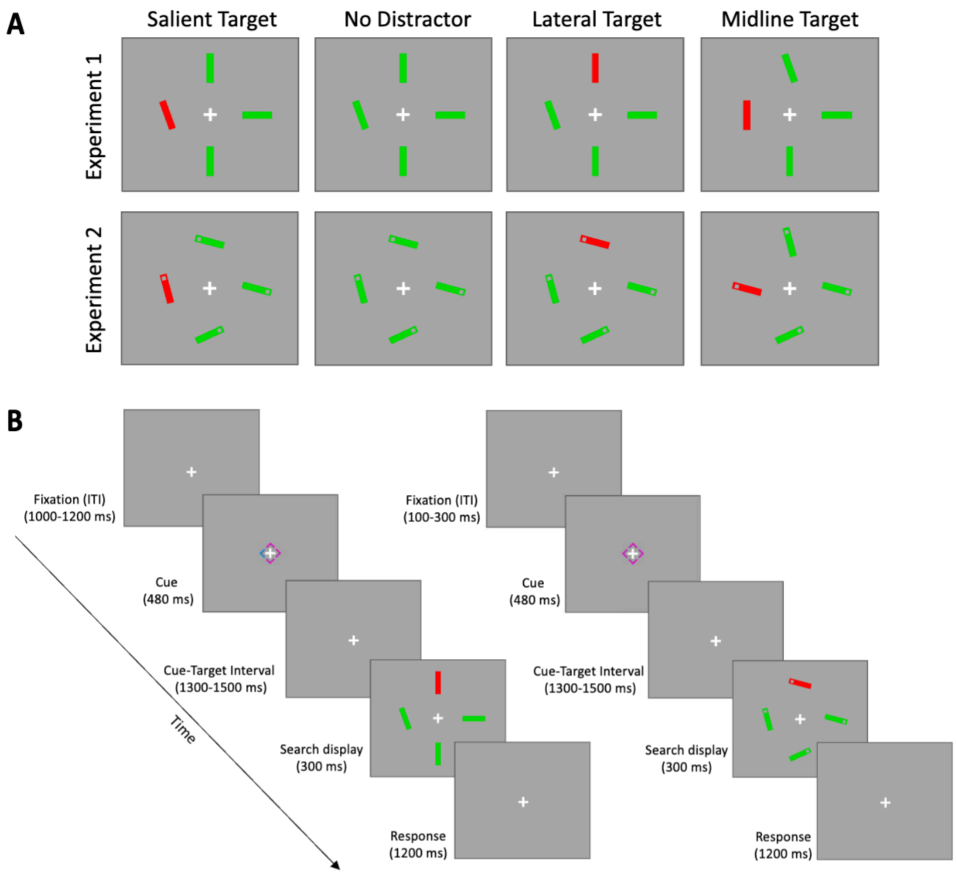

Figure 7. The modified visual search task used in Rashal et al. (2023).

distractor bar, appearing on the vertical midline, was a different color, while the target, on the lateral midline, was the same color as the remaining two bars. The last condition, the midline-target condition, was not included in statistical analyses, as it was the only condition in which the target appeared on the vertical midline. Cues were also standardized across the experiments; before stimuli appeared, a diamond would briefly be presented around the fixation cross. A neutral cue consisted of a magenta diamond, and a valid cue displayed a magenta diamond with one cyan corner. In keeping with the condition design, 75% of valid cues appeared on the lateral midline, while only 25% were on the vertical midline. This is because the target was displayed on the lateral midline in the first three conditions, and only the last condition placed the target on the vertical midline.

The results of the study demonstrated the effects of cues and salient distractors on both components. After performing an analysis of variance (ANOVA) on the EEG data, they showed that salient distractors have a greater effect on the ERP components. In the first experiment, no significance of target salience was found, though higher accuracy following salient targets was seen in the second experiment. The CNV was seen with both neutral and valid cues. Task difficulty had a sizable effect on accuracy, though not on response time. Across both experiments, an effect of cues on the N2pc was seen, since the component reflects attentional direction. A valid cue would fix the participant's attention on the correct target, making the N2pc less pronounced, while a neutral cue would leave the participant without an object of attention, causing them to search the screen for the target. As a result, Rashal et al. (2022) concluded that visual attention is affected by top-down cueing.

With the 2023 iteration, meanwhile, Rashal et al. shifted their focus to neural activity post-stimulus-presentation and examined the SPCN and RLpcN. The design of the two experiments remained largely the same, except for a timing difference discussed below. Since the second experiment was more mentally taxing, Rashal et al. expected to see an effect of task difficulty and target salience on both components. Specifically, they predicted that the amplitude of both components would be greater during the second experiment. In this study, Rashal et al. opted to split the RLpcN into two segments, the first (RLpcN1) corresponding to search difficulty and the second (RLpcN2) corresponding to response preparation. This segmentation of the RLpcN seems somewhat arbitrary, particularly because they divided the component 300 ms after it began, which is typically the time when its amplitude peaks. Within the second experiment, they expected to see a greater effect on the RLpcN1 than the RLpcN2, since the former is correlated with task difficulty.

When statistical analyses were performed on the EEG data, the results largely aligned with their predictions (see Figures 8 and 9). Significant lateralization, as indexed by the SPCN, was seen in all conditions in the second experiment (Rashal et al., 2023). However, for the RLpcN, the second experiment demonstrated a larger effect on the RLpcN2, the response preparation segment. It is possible that this was caused by the increased task difficulty, but it is equally likely that these results can be explained by a timing difference between the first and second experiments. In the first experiment, fixation was jittered after the responses for 1000-1200 ms, while in the second experiment, fixation only lasted between 100 and 300 ms. This timing difference was not present in the 2022 study. Rashal et al. presented their findings without bringing attention to this major difference, which may have acted as a compound and disproportionately affected their results.

The results of this study raise another significant question. Though Rashal et al. operated under the assumption that the RLpcN and the SPCN are two separate components, there is substantial evidence that they represent a single neural process. Though the SPCN is a sharper deflection than the RLpcN, they respond similarly to stimuli and overlap temporally. When considering the graphs in Rashal et al. (2023), the RLpcN2 appears almost identical to the second half of the SPCN, with both gradually falling to a peak of roughly -1.0 ms. The mean scalp area for the components was also observed to be similar, though the SPCN was slightly larger. Splitting this attentional neural process based on whether it is stimulus- or response-

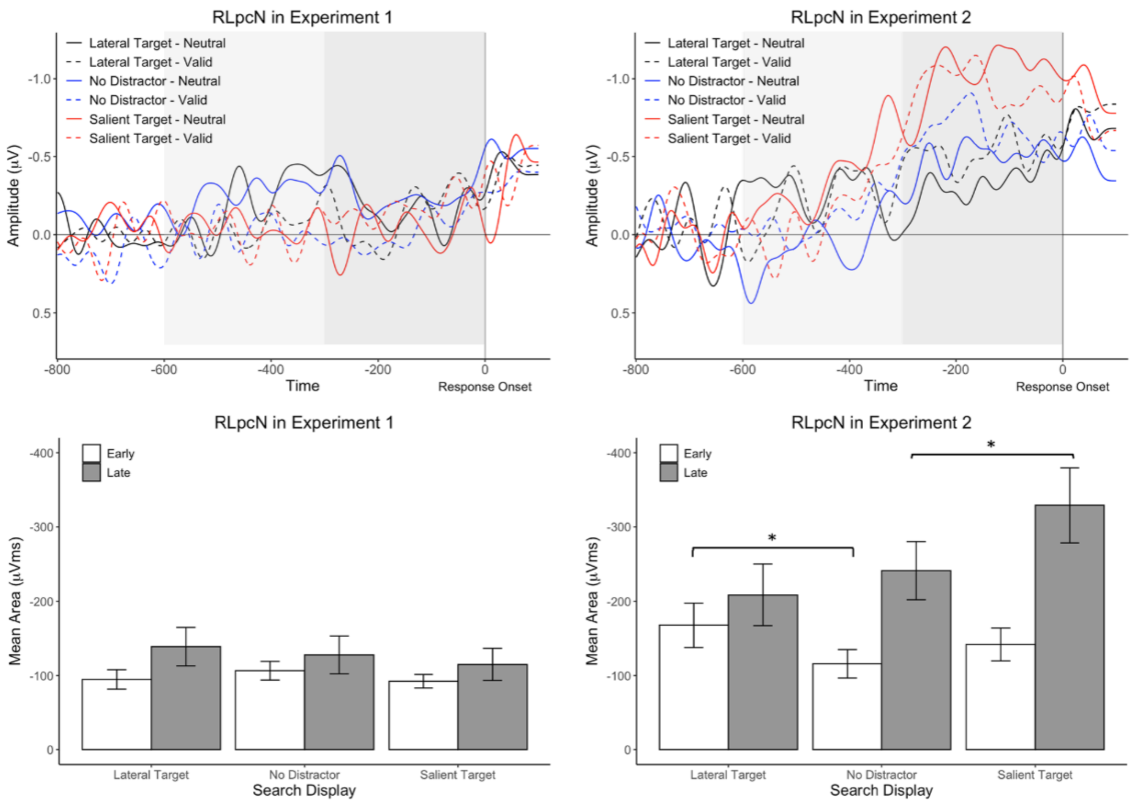

Figure 8. RLpcN data in Rashal et al. (2023).

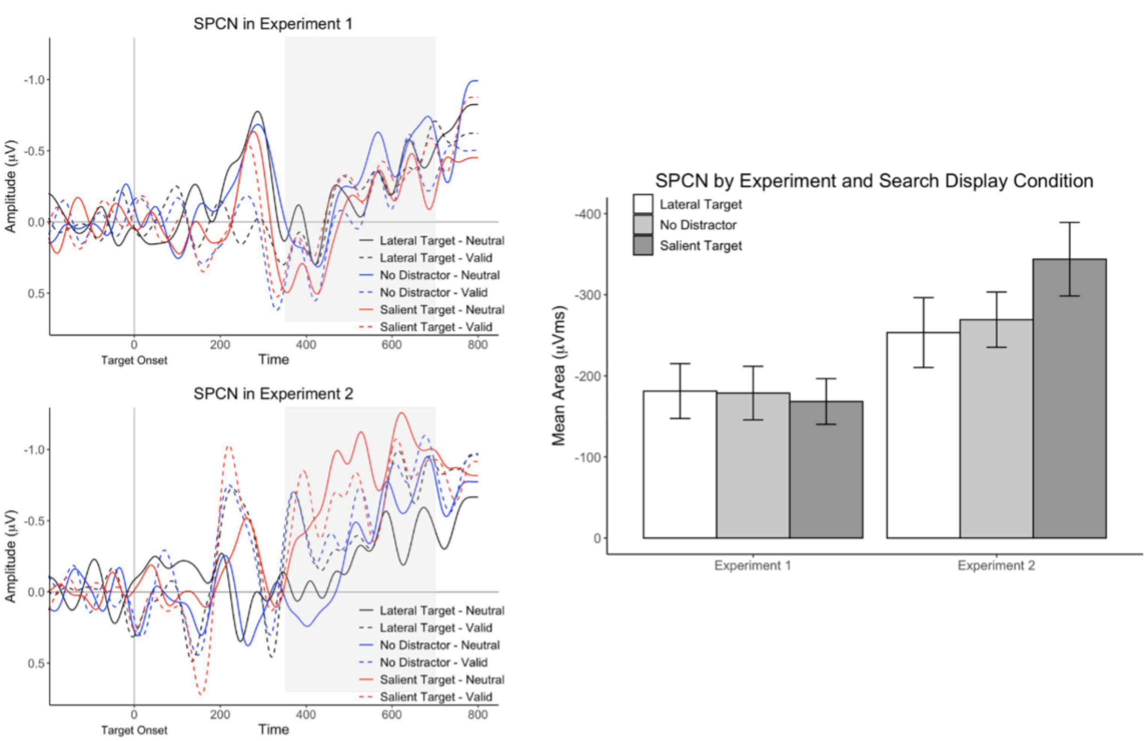

Figure 9. SPCN data in Rashal et al. (2023).

locked can be informative: the SPCN provides information about attentional activity, while the RLpcN captures answer selection and motor preparation. Regardless, further research on the RLpcN will be necessary to determine whether it can be considered its own component.

A recent study by Dolci et al. (2024) modified the visual search task used in Rashal et al (2022; 2023), focusing instead on the interaction between statistical learning and target salience. Since little research has been done on statistical learning, particularly as it relates to the SPCN, Dolci et al. (2024) sought to outline the relationship between statistical learning and visual attention. They retained many aspects of the Rashal et al. task, though they drastically changed the conditions of the experiment, assigning participants to groups based on possible target locations. Within these conditions, targets would be more likely to appear in the "high-frequency" location, though if targets were salient (i.e., targets were red when the distractors were green), they would always appear in the high-frequency location. As with former attentional experiments, Dolci et al. analyzed the SPCN and N2pc, finding main effects of both condition type and target frequency. They chose to segment both components into early and late time windows, and their results showed differences in attentional control in the two segments of the SPCN, while the effects of conditions varied between the segments of the N2pc. The complexity of the modified experiment may have skewed the results, and a simple task that focused solely on statistical learning may have better addressed their initial hypotheses and claims.

Two studies by Heuer & Schubö (2016) and Maheux & Jolicœur (2017) showed similar effects of visual attention on the N2pc and SPCN. Hirai et al. (2022) found that women who had experienced sexual assault had aggravated responses to threatening facial stimuli in a dot-probe task. A behavioral study by Puls & Rothermund (2018), however, examined the role of emotions in facial stimuli with regard to visual attention. The ability of emotional facial stimuli to capture attention has been shown by Holmes et al. (2014), as well as Wójcik et al. (2018), who measured the effects of self-face stimuli. In particular, Holmes et al. (2014) posited that threatening stimuli will attract participants' attention, resulting in an attentional bias. Puls & Rothermund (2018) hypothesized that they would find the opposite effect: emotional facial stimuli have no automatic effect on attention. Several previous reviews (Torrence & Troup, 2018; van Rooijen et al., 2017) indicated mixed results to facial stimuli in EEG data, but Puls & Rothermund chose to perform a behavioral instead of a neurophysiological study, perhaps for convenience purposes.

Table 1. Overview of experimental conditions in all reported studies.

	Experiment 1	Experiment 2	Experiment 3	Experiment 4	Experiment 5	Experiment 6	Experiment 7
n	39	40	39	60	58	39	33
CTI (in ms)	150	150	100 vs. 300	142 vs. 166	24 vs. 150	500 vs. 1000	150
Cue duration (in ms)	150	150	100	24 vs. 48	24 vs. 150	452 vs. 952	150
Cue size (w × h)	512 × 512	512 × 512	256 × 256	128 × 165	128 × 165	512 × 512	256 × 256
Cue face relation	same	different	different	same	same	different	same
Cue shape	rectangular	rectangular	rectangular	oval	oval	rectangular	rectangular
Emotions	angry, happy, fearful	angry, happy, fearful	angry, happy, fearful	angry, happy	angry, happy	angry, happy, fearful	angry, fearful
#Trials/emotion condition	70	120	480	160	160	320	80
Specifics	additional trials with frequency filtered images	additional trials with frequency filtered images		additional trials with frequency filtered images; masked cue presentation (pre- and postmask, 118 ms each)			additional trials with frequency filtered images

Figure 10. Experiments used by Puls & Rothermund (2018).

Their participants (n = 287) completed any of seven experiments (see Figure 10), which all resembled modified dot-probe tasks. Cues were present in all experiments, but their size varied. Stimuli were always human faces, though they varied in emotion: participants could be shown neutral, angry, happy, and fearful faces. The timing and stimulus types changed across the seven experiments, with some including square images and others displaying oval images.

Participants were asked to respond to the orientation or shape of the target stimuli; they were not given feedback. The hypothesis that Puls & Ruthermond put forth was confirmed by their results. Across all seven experiments, none of the emotional facial stimuli produced an attentional capture effect. However, the lack of EEG analysis performed on this data makes the results less compelling, since physiological data could have revealed a different effect than behavioral data.

Materials and Methods

Our current research builds on the recent work done on the SPCN and N2pc. Additionally, we sought to examine how the SPCN and N2pc represent attentional hold and capture, respectively. Sawaki & Luck (2010) defined attentional capture as the ability of a stimulus to quickly and suddenly redirect attention. Attentional hold, conversely, occurs when a stimulus retains attention for a longer duration (Parks & Hopfinger, 2008). Both attentional hold and capture are crucial parts of cognition, allowing people to appropriately assess stimuli in a variety of settings (Hopfinger & Parks, 2012). Many ERP components have been linked to attention (Luck, 2014), but the SPCN has a slow and long-lasting deployment, correlating with attentional hold (Holmes et al., 2009; Wójcik et al., 2018). The N2pc, as stated, has often been coupled with the SPCN in EEG research, and is thought to be associated with attentional capture (Holmes et al., 2009) due to its shorter duration and sharper deflection.

In our experiment, participants (n = 27), all undergraduate students at the University of North Carolina at Chapel Hill, will perform a traditional dot-probe task. All participants will be right-handed, have normal or corrected to normal vision, and have no known neurological disorders. Written and informed consent will be obtained from all participants, as well as a brief demographics and handedness questionnaire. Upon completing the task, participants will receive

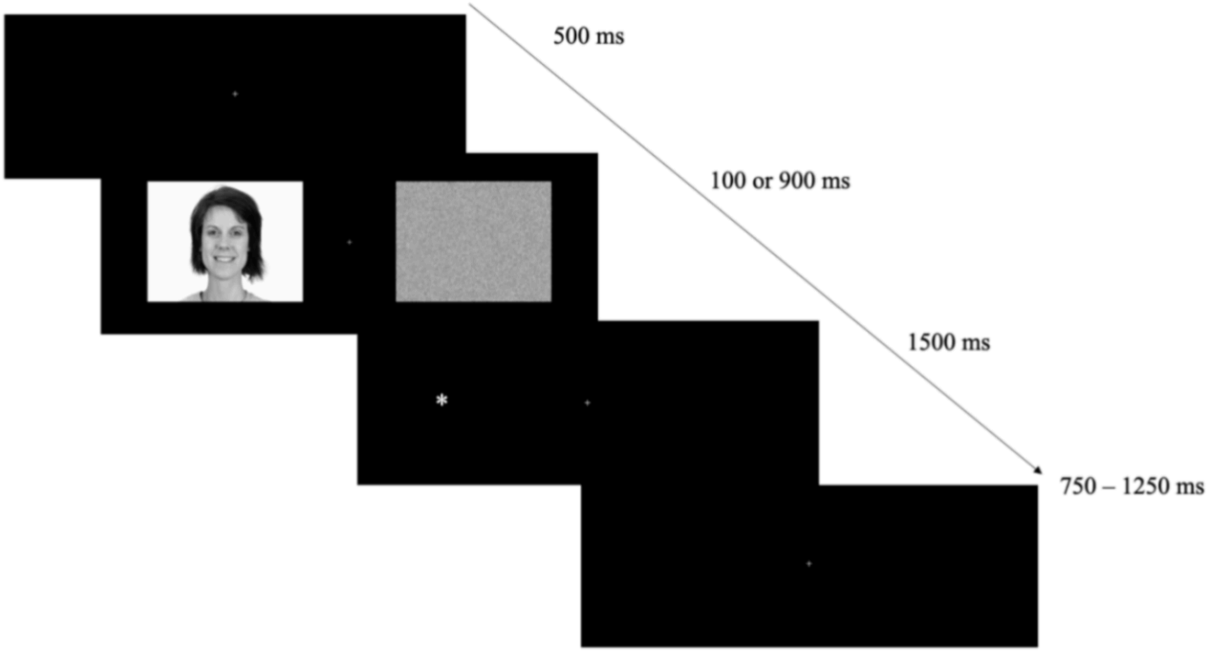

Figure 11. Experiments used in our behavioral research.

participant credit in the SONA app, which will count towards a requirement for PSYC 101, an introductory psychology course at UNC.

Stimuli could be one of three types of images: either "place" images, depicting exteriors of houses and taken from relevant former studies; "face" images, all frontal pictures of happy female faces taken from the Chicago Face Database; or "scrambled" images, created by digitally combining the face and place images. All images were gray-scaled. Exclusively caucasian female faces were used to prevent confounds, while faces with happy expressions have been shown to elicit attentional hold (Torrence et al., 2017). A distinct image was displayed on each side of the fixation cross. Three combinations of the image types were chosen: face and noise, place and noise, and noise and noise. Following stimulus presentation, an asterisk would briefly appear on the screen, either to the left or right of fixation. Participants used the right arrow key to respond when the asterisk appeared right of the lateral midline, and the left arrow key when it appeared left of the lateral midline. The index and middle fingers of the participant's right hand were used for each key; this was done to limit additional motor processing to a minimum.

Within the dot-probe task, we implemented two different stimulus-onset asynchronies (SOA), one shorter (100 ms) and one longer (900 ms). Other than this timing difference, all factors were kept consistent across the experiment. First, the fixation cross would be displayed on a black screen for 500 ms. Stimulus presentation then lasts for either 100 ms or 900 ms. An asterisk then appeared on one side of the screen for 1500 ms or until keypress. Before the next stimulus presentation, an inter-trial interval (ITI) of 750-1250 ms would occur. The duration would be jittered to prevent the participant from anticipating the stimulus. As of yet, we have only performed behavioral pilot experiments, but in future research, we hope to incorporate EEG into our experiment, collecting data on neural activity while participants perform the dot-probe task. We expect to see a strong SPCN and N2pc correlated with attentional hold and capture respectively.

Discussion

Though many studies have investigated the link between visual attention, the SPCN, and the N2pc, many questions have yet to be answered. Our research aims to answer some of those questions, particularly whether the SPCN is a reliable measure of attentional hold, a correlation only proposed by two studies (Holmes et al., 2009; Wójcik et al., 2018). While research has reinforced the importance of the N2pc in visual attention (Chen et al., 2022; Dolci et al., 2024; Drisdelle & Jolicœur, 2019; Heuer & Schubö, 2016; Maheux & Jolicœur, 2017; Rashal et al., 2022), the SPCN is still a relatively new component, and there have been a number of hypotheses about its true neural source. Many of the articles discussed in this paper use visual search tasks to analyze the SPCN, but we aim to take a slightly different approach with the dot-probe task, used by Holmes et al. (2009) and Wójcik et al. (2018) which relies more on attention than working memory. As research on visual attention continues, we hope to see stronger correlations between the SPCN and attentional hold.

If the mechanisms of the SPCN are not fully understood, the RLpcN, identified by Drisdelle & Jolicœur (2019), is even less clearly defined. Rashal et al. (2023) assert that the RLpcN is its own valid component, stating that it is correlated with response preparation, as opposed to the SPCN, associated with attentional selection. However, their EEG data shows that the RLpcN is a near-exact replica of the SPCN, calling their assertion into question (Rashal et al., 2023). In the future, more research on the RLpcN and SPCN is necessary to determine the interplay and overlap between these two components.

The three components highlighted in this paper, the N2pc, RLpcN, and SPCN, illustrate the brain's mechanisms of facilitating attention, and continued investigation into these components is crucial. Different approaches, both physiological and behavioral, will provide greater context on the nuances of attention. Both visual search tasks and dot-probe tasks have proved useful when examining visual attention, but incorporating other experimental measures in future research, such as working memory tests, may provide additional information. Visual attention influences almost every aspect of human life (Hopfinger et al., 2000), with attentional hold and capture enriching cognition and perception (Hopfinger & Parks, 2012). Further exploration of attention and the associated neural processes will enhance our understanding of the way we navigate the world.

References

Chen, Y., Brigadoi, S., Schiano Lomoriello, A., Jolicœur, P., Simal, A., Fu, S., Baro, V., & Dell'Acqua, R. (2022). A bilateral SPCN is elicited by to-be-memorized visual stimuli displayed along the vertical midline. *Psychophysiology*, *59*(8), e14045. https://doi.org/10.1111/psyp.14045.

Dolci, C., Rashal, E., Santandrea, E., Ben Hamed, S., Chelazzi, L., Macaluso, E., & Boehler, C. N. (2024). The dynamics of statistical learning in visual search and its interaction with salience processing: An EEG study. *NeuroImage*, *286*, 120514. https://doi.org/10.1016/j.neuroimage.2024.120514.

Drisdelle, B. L., & Jolicœur, P. (2019). Stimulus- and Response-locked Posterior Contralateral Negativity Bisect Cognitive Operations in Visual Search. *Journal of Cognitive Neuroscience*, *31*(4), 574–591. https://doi.org/10.1162/jocn_a_01364.

Eimer, M., & Kiss, M. (2007). Attentional capture by task-irrelevant fearful faces is revealed by the N2pc component. *Biological Psychology*, *74*(1), 108–112. https://doi.org/10.1016/j.biopsycho.2006.06.008.

Heuer, A., & Schubö, A. (2016). The Focus of Attention in Visual Working Memory: Protection of Focused Representations and Its Individual Variation. *PLOS ONE*, *11*(4), e0154228. https://doi.org/10.1371/journal.pone.0154228.

Hickey, C., Di Lollo, V., & McDonald, J. J. (2009). Electrophysiological Indices of Target and Distractor Processing in Visual Search. *Journal of Cognitive Neuroscience*, *21*(4), 760–775. https://doi.org/10.1162/jocn.2009.21039.

Hirai, M., Hernandez, E. N., Villarreal, D. Y., & Clum, G. A. (2022). Attentional Bias Toward Threat in Sexually Victimized Hispanic Women: A Dot Probe Study. *Journal of Trauma & Dissociation*, *23*(1), 110-123. https://doi.org/10.1080/15299732.2021.1989108.

Holmes, A., Bradley, B. P., Kragh Nielsen, M., & Mogg, K. (2009). Attentional selectivity for emotional faces: Evidence from human electrophysiology. *Psychophysiology*, *46*(1), 62-68. https://doi.org/10.1111/j.1469-8986.2008.00750.x.

Holmes, A., Mogg, K., de Fockert, J., Nielsen, M. K., & Bradley, B. P. (2014). Electrophysiological evidence for greater attention to threat when cognitive control resources are depleted. *Cognitive, Affective, & Behavioral Neuroscience*, *14*(2), 827–835. https://doi.org/10.3758/s13415-013-0212-4.

Hopfinger, J. B., Buonocore, M. H., & Mangun, G. R. (2000). The neural mechanisms of top-down attentional control. *Nature Neuroscience*, *3*(3), 284–291. https://doi.org/10.1038/72999.

Hopfinger, J. B., & Parks, E. L. (2012). Involuntary attention. In G. R. Mangun (Ed.), *The neuroscience of attention: Attentional control and selection* (pp. 30–53). Oxford University Press. https://psycnet.apa.org/doi/10.1093/acprof:oso/9780195334364.003.0002.

Jolicœur, P., Sessa, P., Dell'Acqua, R., et al. On the control of visual spatial attention: evidence from human electrophysiology. *Psychological Research*, *70*, 414–424 (2006). https://doi.org/10.1007/s00426-005-0008-4.

Luck, S. J. (1993). *The role of selective attention in the perception of multiple-element visual arrays: Cognitive and neural mechanisms.* University of California, San Diego.

Luck, S. J. (2014). *An introduction to the event-related potential technique*. MIT press.

Luck, S. J., & Hillyard, S. A. (1994). Spatial filtering during visual search: Evidence from human electrophysiology. *Journal of Experimental Psychology: Human Perception and Performance*, *20*(5), 1000–1014. https://doi.org/10.1037/0096-1523.20.5.1000.

Luck, S. J., & Kappenman, E. S. (2012). *The Oxford handbook of event-related potential components*. Oxford University Press.

Parks, E. L., & Hopfinger, J. B. (2008). Hold it! Memory affects attentional dwell time. *Psychonomic bulletin & review*, *15*(6), 1128-1134. https://doi.org/10.3758/PBR.15.6.1128.

Maheux, M., & Jolicœur, P. (2017). Differential engagement of attention and visual working memory in the representation and evaluation of the number of relevant targets and their spatial relations: Evidence from the N2pc and SPCN. *Biological Psychology*, *125*, 28–35. https://doi.org/10.1016/j.biopsycho.2017.01.011.

Puls, S., & Rothermund, K. (2018). Attending to emotional expressions: No evidence for automatic capture in the dot-probe task. *Cognition and Emotion*, *32*(3), 450–463. https://doi.org/10.1080/02699931.2017.1314932.

Rashal, E., Senoussi, M., Santandrea, E., Ben-Hamed, S., Macaluso, E., Chelazzi, L., & Boehler, C. N. (2022). An EEG study of the combined effects of top-down and bottom-up attentional selection under varying task difficulty. *Psychophysiology*, *59*(6), e14002. https://doi.org/10.1111/psyp.14002.

Rashal, E., Santandrea, E., Ben-Hamed, S., Macaluso, E., Chelazzi, L., & Boehler, C. N. (2023). Effects of top-down and bottom-up attention on post-selection posterior contralateral negativity. *Attention, Perception, & Psychophysics*, *85*(3), 705–717. https://doi.org/10.3758/s13414-022-02636-7.

Sawaki, R., & Luck, S. J. (2010). Capture versus suppression of attention by salient singletons: Electrophysiological evidence for an automatic attend-to-me signal. *Attention, Perception, & Psychophysics*, *72*(6), 1455-1470. https://doi.org/10.3758/app.72.6.1455.

Torrence, R. D., & Troup, L. J. (2018). Event-related potentials of attentional bias toward faces in the dot-probe task: A systematic review. *Psychophysiology*, *55*(6), e13051. https://doi.org/10.1111/psyp.13051.

Torrence, R. D., Wylie, E., & Carlson, J. M. (2017). The time-course for the capture and hold of visuospatial attention by fearful and happy faces. *Journal of Nonverbal Behavior*, *41*(2), 139- 153. https://doi.org/10.1007/s10919-016-0247-7.

van Rooijen, R., Ploeger, A., & Kret, M. E. (2017). The dot-probe task to measure emotional attention: A suitable measure in comparative studies? *Psychonomic Bulletin & Review*, *24*(6), 1686–1717. https://doi.org/10.3758/s13423-016-1224-1.

Wójcik, M. J., Nowicka, M. M., Kotlewska, I., & Nowicka, A. (2018). Self-face Captures, Holds, and Biases Attention. *Frontiers in Psychology*, *8*. https://doi.org/10.3389/fpsyg.2017.02371.